Lattice Yang–Mills theory at strong coupling III: confinement (area law)

A sequel to the foundations monograph of the project "Solving Yang–Mills and the Mass Gap"

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Abstract

We prove Wilson's area law for lattice gauge theories at strong coupling, with complete proofs and explicit constants, in the framework of the foundations monograph [F] of this project. For $G = SU(N)$, every dimension $d \geq 2$, every periodic lattice with all sides $L_\lambda \geq 3$, every $0 < \beta < \beta_0(d)$ (a threshold below which the cluster expansion of [F] converges; it does not depend on N), every rectangular $R \times T$ loop C in a coordinate plane (μ, ν) and every irreducible representation r of non-zero N-ality,

$$\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \leq d_r \exp\big(-\sigma(\beta)A + \kappa(\beta)|C|\big), \qquad A = \min(RT,\, L_\mu L_\nu - RT),$$

with $\sigma(\beta) = \log\big(1/(e^2 K_{\mathcal{P}}(e^{2\beta}-1))\big) \to \infty$ as $\beta \downarrow 0$, $K_{\mathcal{P}} = 4(2d-3)$, and an explicit $\kappa(\beta) \leq 21/20$. Consequently the string tension of every infinite-volume limit point of the periodic Wilson measures is at least $\sigma(\beta)$. The proof combines a centre (charge) selection rule, a projection chain map onto a coordinate 2-torus together with the $\mathbb{Z}_N$-homology of the 2-torus (which produces the torus area A), the polymer expansion of [F], and a self-contained inductive bound for ratios of polymer partition functions. Only a central element of finite order acting non-trivially in r is used. The theorem therefore holds verbatim for $U(1)$ (every non-trivial representation) and for $U(N)$ (every representation that is non-trivial on the centre); for $U(1)$ in four dimensions the area law fails at weak coupling (proved for the Villain action, and asserted for Wilson's action, by Fröhlich and Spencer), so strong-coupling confinement is not a non-abelian phenomenon. We also prove that in $d = 2$ the area law holds for every non-trivial representation, that every rectangular Wilson loop in a non-trivial representation decays at least exponentially in its perimeter, and that loops winding around the torus with non-zero charge have vanishing expectation; and we compare with Osterwalder–Seiler (1978) and Seiler (1982), where the strong-coupling area law was first proved.

Contents

1 Introduction and main result **2**

2 Setting **3**
2.1 Standing assumptions and facts from [F] 3
2.2 Closed paths, Wilson loops and the charge of a representation 4
2.3 The constants 5

3 Chains on the torus and the area of a loop **6**

4 The centre selection rule **7**

5 Expansion of a Wilson loop and the ratio of partition functions **9**

6 The area law **12**

7 Infinite volume and the string tension **14**

8 Representations without central charge; two dimensions **15**

9 Discussion **16**

1 Introduction and main result

The foundations monograph [F] develops lattice Yang–Mills theory on a periodic lattice: Wilson's measure, its gauge invariance, reflection positivity, the transfer matrix, the reduction of the continuum problem to lattice hypotheses, a volume-uniform mass gap for $\beta < \beta_D(d)$, and the polymer (closed-surface) representation of the measure with a convergent cluster expansion for $\beta < \beta_0(d)$. Among the consequences of the latter which are listed but not proved there ([F, Remark 'Further consequences, not proved here' (`rem:further`)]) is Wilson's area law for Wilson loops in representations of non-trivial N-ality. This article supplies that proof, in the notation and with the conventions of [F], and with explicit constants.

Wilson's criterion [Wil74] says that static charges in a representation r are confined if the Wilson loop expectation of large rectangular loops decays exponentially in the *area* spanned by the loop, which (via the transfer matrix) means that the energy of a static charge–anticharge pair grows linearly with their distance. The area law at small β was proved by Osterwalder and Seiler [OS78] and, for general loops, by Seiler [Sei82]; see Remark 9.2 for a comparison.

Theorem 1.1 (Main theorem: area law for $SU(N)$ at strong coupling)**.** *Let $G = SU(N)$, $N \geq 2$, with the Wilson measure $\langle\cdot\rangle_{\Lambda,\beta}$ of [F, Definition 'Wilson measure' (`def:wilson`)] (ρ the defining representation). Let $d \geq 2$, let Λ be a periodic lattice with all $L_\lambda \geq 3$, and let*

$$0 < \beta < \beta_0(d) = \tfrac{1}{2}\log\Big(1 + \frac{1}{e^2 K_{\mathcal{P}}(K_{\mathcal{P}}+2)}\Big), \qquad K_{\mathcal{P}} = 4(2d-3).$$

Let C be a rectangular loop with sides R and T in a coordinate plane (μ,ν) of Λ, where $1 \leq R \leq L_\mu - 1$ and $1 \leq T \leq L_\nu - 1$, let $|C| = 2(R+T)$ be its length and $A := \min(RT,\, L_\mu L_\nu - RT)$, and let r be an irreducible representation of $SU(N)$ whose N-ality $n(r)$ is not divisible by N. Then

$$\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \leq d_r \exp\big(-\sigma(\beta)\, A + \kappa(\beta)\, |C|\big), \tag{1}$$

where, with $x(\beta) := e^2 K_{\mathcal{P}}\,(e^{2\beta}-1) \in (0, \frac{1}{K_{\mathcal{P}}+2})$,

$$\sigma(\beta) = \log\frac{1}{x(\beta)}, \qquad \kappa(\beta) = 2(d-1)\Big[\log\Big(1+\frac{1}{K_{\mathcal{P}}}\Big) + \frac{x(\beta)}{1-x(\beta)}\Big] + \frac{1}{4}\log\frac{1}{1-x(\beta)}. \tag{2}$$

Moreover $\sigma(\beta) > \log(K_{\mathcal{P}}+2)$, $\sigma(\beta) = \log\frac{1}{2e^2 K_{\mathcal{P}}\beta} + O(\beta) \to \infty$ as $\beta \downarrow 0$, and $\kappa(\beta) < \frac{d-1}{2d-3} + \frac{1}{4(K_{\mathcal{P}}+1)} \leq \frac{21}{20}$.

Since reversing a loop replaces $\chi_r(U_C)$ by its complex conjugate, cyclic shifts of C do not change it, and the Wilson measure is translation invariant ([F, Theorem 'Basic properties' (`thm:basic`)](iv)), the bound (1) holds for every rectangle with sides R, T in a coordinate plane, whatever its base point, starting point and orientation.

Theorem 1.1 is Corollary 6.4 below. The threshold $\beta_0(d)$ is the one below which [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)] proves convergence of the cluster expansion (a lower bound for its radius of convergence, not the radius itself); in the normalisation $w_\beta = \exp(\frac{\beta}{N}\mathrm{Re}\,\mathrm{Tr}\,U)$ of [F] it does not depend on N. Numerically, $\beta_0(2) \approx 2.81 \cdot 10^{-3}$, $\beta_0(3) \approx 4.03 \cdot 10^{-4}$, $\beta_0(4) \approx 1.54 \cdot 10^{-4}$. The constants are far from optimal (Remark 9.3).

Further results. The proof gives more than Theorem 1.1:

(i) a general form (Theorem 6.2) for any compact group G as in [F] with a central element z_0 of finite order M acting non-trivially in r, and any closed lattice path, the area being replaced by the minimal size of a $\mathbb{Z}_M$-valued 2-chain with the prescribed boundary;

(ii) the same area law for $U(1)$, for every non-trivial representation, and for $U(N)$, for every representation non-trivial on the centre (Corollary 6.5); this is the precise sense in which strong-coupling confinement fails the "$U(1)$ test" of the research program (Remark 9.1);

(iii) a lower bound $\sigma(\beta)$ on the string tension of every infinite-volume limit point of the periodic Wilson measures (Corollary 7.2);

(iv) a variant in which the perimeter coefficient tends to 0 as $\beta \downarrow 0$ (Corollary 6.6);

(v) in $d = 2$, the area law for *every* non-trivial representation of any G (Proposition 8.4);

(vi) for rectangular loops in every non-trivial representation, in every d and for all $\beta < \beta_0(d)$, decay at least exponential in the perimeter (Proposition 8.2), and the vanishing of loops which wind around the torus with non-zero charge (Proposition 4.5).

Strategy. Write $w_\beta = c_0(\beta)(1 + f_\beta)$ and expand $\prod_p(1 + f_\beta(U_p))$ as in [F]. (1) Grouping the plaquettes connected to the loop C into one "loop polymer" X, the numerator of $\langle \chi_r(U_C) \rangle$ becomes $\sum_X \Phi(X)\, \Xi(\text{plaquettes avoiding } X \cup C)$, where $\Phi(X)$ is the integral of $\chi_r(U_C) \prod_{p \in X} f_\beta(U_p)$ (Lemma 5.3). (2) Decomposing f_β according to the action of the centre, a single-link substitution $U_\ell \mapsto z_0 U_\ell$ shows that $\Phi(X)$ is a sum of terms indexed by $\mathbb{Z}_M$-valued 2-chains ζ supported in X, each of which vanishes unless $\partial\zeta = -m(r)\,[C]$ (Lemma 4.3). (3) Projecting onto the coordinate 2-torus of the loop by a chain map and using that the 2-cycles of a 2-torus are the multiples of its fundamental class, such a ζ has at least A plaquettes (Proposition 3.7). (4) The ratio of polymer partition functions is at most $e^{\theta(|X|(K_{\mathcal{P}}+1)+2(d-1)|E_C|)}$ by an inductive lemma of Kotecký–Preiss–Dobrushin type (Lemma 5.4) whose hypothesis is the estimate proved in [F]. (5) Connected sets attached to C are counted with a rooted tree encoding (Lemma 5.8), which produces the perimeter term.

What is used from [F]. The definitions of [F] and the following results, whose proofs we have re-checked: Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`) (standard facts), the bounds (3) (from (`eq:w`), `cor:uniform` and (`eq:eps`)), Proposition 'Polymer representation' (`prop:polymer`), Lemma 'Closed surfaces' (`lem:closed`), Lemma 'Counting polymers' (`lem:count`) and the estimate in the proof of Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`). For the bound $c_0(\beta) \geq 1$ we give a slightly completed argument (Lemma 2.2(b)). The cited Kotecký–Preiss theorem [F, Cited Theorem 'Kotecký–Preiss 1986' (`thm:KP`)] is *not* used: Lemma 5.4 gives what is needed with a complete proof. Labels follow [F, Convention 'Labels']: every Theorem, Proposition, Lemma and Corollary below has a complete proof.

2 Setting

2.1 Standing assumptions and facts from [F]

Convention 2.1 (Standing assumptions). G, $\rho : G \to U(N)$, $\operatorname{Tr} U := \operatorname{Tr} \rho(U)$, dU and r, d_r, D^r, χ_r are as in [F, Convention 'Group']. We fix $d \geq 2$ and the periodic lattice $\Lambda = \prod_{\lambda=0}^{d-1} \mathbb{Z}/L_\lambda\mathbb{Z}$ with

$$L_\lambda \geq 3 \qquad (0 \leq \lambda \leq d-1),$$

with links $\ell = (x, \lambda)$, plaquettes $p = (x, \lambda\lambda')$ $(\lambda < \lambda')$, the sets $E(\Lambda)$ and $\mathcal{P} := \mathcal{P}(\Lambda)$, configurations $U \in \Omega_\Lambda = G^{E(\Lambda)}$ with product Haar measure dU, and plaquette variables U_p as in [F, Convention 'Lattice'], [F, Convention 'Configurations'] and [F, eq. `eq:plaquette`]. For a plaquette p, $E(p)$ is the set of its four links (they are distinct, since $L_\lambda \geq 2$), and $E(X) := \bigcup_{p \in X} E(p)$ for $X \subseteq \mathcal{P}$. Distinct plaquettes are *adjacent* if they share a link; connectedness and connected components of sets of plaquettes refer to this adjacency. The Wilson measure, its expectation $\langle \cdot \rangle_{\Lambda,\beta}$ and its partition function $Z_\Lambda(\beta)$ are those of [F, Definition 'Wilson measure' (`def:wilson`)], with plaquette weight $w_\beta(U) = \exp(\frac{\beta}{N} \operatorname{Re} \operatorname{Tr} U)$ and character coefficients $c_r(\beta) = \int \overline{\chi_r}\, w_\beta \, dU$ ([F, Definition 'Plaquette weight']). As in [F, Definition `def:polymer`] we put $f_\beta := w_\beta / c_0(\beta) - 1$ and $\epsilon(\beta) := e^{2\beta} - 1$; a polymer is a non-empty connected set $\gamma \subseteq \mathcal{P}$, polymers γ, γ' are compatible ($\gamma \approx \gamma'$) iff $E(\gamma) \cap E(\gamma') = \emptyset$, and the activity of γ is $z(\gamma) = \int \prod_{p \in \gamma} f_\beta(U_p)\, dU$. We write $K := K_{\mathcal{P}} = 4(2d-3)$.

Lemma 2.2 (Facts from [F])**.** *For every $\beta \geq 0$:*

(a) w_β is a continuous class function with $w_\beta(U^{-1}) = w_\beta(U)$ and $e^{-\beta} \leq w_\beta \leq e^{\beta}$;

(b) $1 \leq c_0(\beta) \leq e^{\beta}$; hence f_β is real and continuous, $1 + f_\beta = w_\beta / c_0(\beta) > 0$, $\int f_\beta \, dU = 0$, and

$$\|f_\beta\|_\infty \leq \epsilon(\beta) = e^{2\beta} - 1; \tag{3}$$

(c) $\int f_\beta(AU^{\pm 1}B)\, dU = 0$ for all $A, B \in G$;

(d) every link lies in exactly $2(d-1)$ plaquettes, and every plaquette is adjacent to at most K plaquettes;

(e) $|z(\gamma)| \leq \epsilon(\beta)^{|\gamma|}$ for every polymer γ;

(f) for every plaquette p_0 and $n \geq 1$, the number of polymers $\gamma \ni p_0$ with $|\gamma| = n$ is at most $(eK)^n$.

Proof. (a) is [F, eq. `eq:w`]: $\rho(U)$ is unitary, so $|\operatorname{Re} \operatorname{Tr} U| \leq N$, and $\operatorname{Tr} \rho(gUg^{-1}) = \operatorname{Tr} \rho(U)$, $\operatorname{Tr} \rho(U^{-1}) = \overline{\operatorname{Tr} \rho(U)}$. (b) $c_0 \leq e^{\beta}$ by (a). By Jensen's inequality, $c_0 = \int e^{\frac{\beta}{N} \operatorname{Re} \operatorname{Tr} \rho(U)}\, dU \geq \exp(\frac{\beta}{N} \operatorname{Re} \int \operatorname{Tr} \rho(U)\, dU)$. Decomposing ρ into irreducible representations and using $\int D^{r'}_{aa}\, dU = 0$ for $r' \neq 0$ ([F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](ii) with the trivial representation), $\int \operatorname{Tr} \rho(U)\, dU$ is the multiplicity of the trivial representation in ρ, a non-negative integer; hence $c_0 \geq 1$. (This is the argument of [F, Corollary `cor:uniform`], where $\int \operatorname{Tr} U \, dU = 0$ is used; that holds if ρ has no trivial summand, e.g. for $SU(N)$ and $U(1)$, and in general the integral is ≥ 0, which suffices.) Then $w_\beta / c_0 \in [e^{-2\beta}, e^{\beta}]$, so $f_\beta \in [e^{-2\beta} - 1, e^{\beta} - 1]$, which gives (3) because $1 - e^{-2\beta} \leq e^{2\beta} - 1$; and $\int f_\beta = c_0/c_0 - 1 = 0$. (c) By left, right and inversion invariance of dU ([F, Cited Theorem `thm:pw`](i)), $\int f_\beta(AU^{\pm 1}B)\, dU = \int f_\beta \, dU = 0$. (d) The plaquettes containing (x, λ) are, for each $\lambda' \neq \lambda$, the two plaquettes of the plane $\{\lambda, \lambda'\}$ based at x and at $x - e_{\lambda'}$; these $2(d-1)$ plaquettes are distinct because $L_{\lambda'} \geq 2$. A plaquette has four links, each lying in $2d-3$ further plaquettes, whence at most $4(2d-3) = K$ adjacent plaquettes. (e) is (3). (f) is [F, Lemma 'Counting polymers' (`lem:count`)]; it is also the special case $J = K$, root $o = p_0$, of Lemma 5.8 below, which gives the bound $\binom{Kn}{n-1} \leq K^{n-1}(1 + \frac{1}{K})^{Kn} \leq (eK)^n$. □

2.2 Closed paths, Wilson loops and the charge of a representation

Definition 2.3 (Closed paths, Wilson loops, rectangles)**.** An oriented link is a pair ℓ^ϵ, $\ell = (x, \lambda) \in E(\Lambda)$, $\epsilon \in \{\pm 1\}$, running from x to $x + e_\lambda$ if $\epsilon = 1$ and from $x + e_\lambda$ to x if $\epsilon = -1$. A *closed path* is a finite sequence $C = (\ell_1^{\epsilon_1}, \ldots, \ell_n^{\epsilon_n})$, $n \geq 1$, of oriented links, each ending where the next one starts, and the last ending where the first starts. Its length is $|C| := n$, its set of

links $E_C := \{\ell_1, \dots, \ell_n\}$, and its holonomy $U_C := U_{\ell_1}^{\epsilon_1} U_{\ell_2}^{\epsilon_2} \cdots U_{\ell_n}^{\epsilon_n}$. For irreducible r, $\chi_r(U_C)$ is a continuous function on Ω_Λ with $|\chi_r(U_C)| \le d_r$, unchanged under cyclic shifts of C; reversing C replaces it by $\overline{\chi_r(U_C)}$.

A *rectangular loop* is specified by directions $\mu < \nu$, a base point $x^0 \in \Lambda$ and integers $1 \le R \le L_\mu - 1$, $1 \le T \le L_\nu - 1$. With $x_{ij} := x^0 + ie_\mu + je_\nu$ it is the closed path C which traverses (x_{i0}, μ) forwards for $i = 0, \dots, R-1$, then (x_{Rj}, ν) forwards for $j = 0, \dots, T-1$, then (x_{iT}, μ) backwards for $i = R-1, \dots, 0$, then (x_{0j}, ν) backwards for $j = T-1, \dots, 0$. Because $R < L_\mu$ and $T < L_\nu$, its $2(R+T)$ links are distinct, so $|C| = |E_C| = 2(R+T)$, and C visits $2(R+T)$ distinct sites, each lying on exactly two links of E_C. We write

$$D := \{(x_{ij}, \mu\nu) : 0 \le i < R,\ 0 \le j < T\}, \qquad \Sigma := \{(x, \mu\nu) : x_\lambda = x^0_\lambda \text{ for all } \lambda \notin \{\mu, \nu\}\},$$

($|D| = RT$, $|\Sigma| = L_\mu L_\nu$, $D \subseteq \Sigma$), $\mathcal{P}_{\mu\nu} := \{(x, \mu\nu) : x \in \Lambda\}$ for the plaquettes parallel to the plane of C, and

$$A := \min(RT,\ L_\mu L_\nu - RT)\ \ge 1.$$

Lemma 2.4 (Central elements act by scalars)**.** *Let $z_0 \in G$ be central with $z_0^M = 1$ for some $M \ge 1$, and $\omega := e^{2\pi i/M}$. For every irreducible representation r there is a unique $m(r) \in \mathbb{Z}_M := \mathbb{Z}/M\mathbb{Z}$ with $D^r(z_0) = \omega^{m(r)}\mathbb{1}$. Consequently $\chi_r(z_0^j U) = \omega^{j\,m(r)}\chi_r(U)$ for all $j \in \mathbb{Z}$ and $U \in G$.*

Proof. $D^r(z_0)$ commutes with every $D^r(g)$. If c is an eigenvalue of $D^r(z_0)$, the eigenspace $\ker(D^r(z_0) - c)$ is non-zero and invariant under $D^r(G)$, hence equal to the whole space by irreducibility, i.e. $D^r(z_0) = c\mathbb{1}$ (Schur's lemma). Since $D^r(z_0)^M = D^r(z_0^M) = \mathbb{1}$, $c^M = 1$, i.e. $c = \omega^m$ for a unique $m \in \mathbb{Z}_M$. Finally $\chi_r(z_0^j U) = \mathrm{Tr}(D^r(z_0)^j D^r(U)) = \omega^{jm}\chi_r(U)$. □

Definition 2.5 (Charge, N-ality)**.** Given z_0 as in Lemma 2.4, $m(r) \in \mathbb{Z}_M$ is the *z_0-charge* of r. For $G = SU(N)$ we take $z_0 = e^{2\pi i/N}\mathbb{1}_N$ (a generator of the centre $\mathbb{Z}_N$), $M = N$, and call $n(r) := m(r) \in \mathbb{Z}_N$ the *N-ality* of r.

Remark 2.6. If r occurs in $\rho^{\otimes a} \otimes \bar\rho^{\otimes b}$ then $n(r) = a - b \bmod N$, since $\rho(z_0) = \omega\mathbb{1}$ acts on this tensor product by ω^{a-b}; every irreducible representation of $SU(N)$ occurs in such a product (see the proof of [F, Lemma 'Positivity of the character coefficients' (`lem:cr`)]). For instance the fundamental, the antifundamental and the adjoint representation have N-ality 1, $N-1$ and 0; for highest weight λ, $n(r)$ is the number of boxes of the Young diagram modulo N. None of this is used below.

2.3 The constants

Definition 2.7 (Constants)**.** For $\beta \ge 0$ with $x(\beta) < 1$ put

$$x(\beta) := e^2 K \epsilon(\beta), \qquad \theta(\beta) := \frac{x(\beta)}{1 - x(\beta)}, \qquad \sigma(\beta) := \log\frac{1}{x(\beta)},$$

$$\kappa_1(\beta) := 2(d-1)\Big[\log\Big(1 + \frac{1}{K}\Big) + \theta(\beta)\Big], \qquad \kappa(\beta) := \kappa_1(\beta) + \frac{1}{4}\log\frac{1}{1 - x(\beta)},$$

and recall $\beta_0(d) = \frac{1}{2}\log(1 + \frac{1}{e^2 K(K+2)})$ from [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)].

Lemma 2.8. *Let $0 < \beta < \beta_0(d)$. Then $0 < x(\beta) < \frac{1}{K+2}$, $(K+1)\theta(\beta) < 1$, $\sigma(\beta) > \log(K+2)$, $\kappa_1(\beta) < \frac{d-1}{2d-3}$ and $\kappa(\beta) < \frac{d-1}{2d-3} + \frac{1}{4(K+1)} \le \frac{21}{20}$. As $\beta \downarrow 0$, $\sigma(\beta) = \log\frac{1}{2e^2 K\beta} + O(\beta)$ and $\kappa(\beta) \to 2(d-1)\log(1 + \frac{1}{K})$.*

Proof. $\beta < \beta_0(d)$ is equivalent to $\epsilon(\beta) < \frac{1}{e^2K(K+2)}$, i.e. to $x < \frac{1}{K+2}$; then $\theta < \frac{1/(K+2)}{(K+1)/(K+2)} = \frac{1}{K+1}$ and $\sigma > \log(K+2)$. Using $\log(1+\frac{1}{K}) < \frac{1}{K}$ and $\theta < \frac{1}{K+1} < \frac{1}{K}$, $\kappa_1 < \frac{4(d-1)}{K} = \frac{d-1}{2d-3}$, and $\frac{1}{4}\log\frac{1}{1-x} < \frac{1}{4}\log\frac{K+2}{K+1} < \frac{1}{4(K+1)}$. Since $d \geq 2$, $\frac{d-1}{2d-3} \leq 1$ and $K+1 \geq 5$. Finally $\epsilon(\beta) = 2\beta(1+O(\beta))$ and $x, \theta \to 0$. □

3 Chains on the torus and the area of a loop

Definition 3.1 (Cellular chains)**.** Let $\mathbb{A}$ be an abelian group (we use $\mathbb{A} = \mathbb{Z}$ and $\mathbb{A} = \mathbb{Z}_M$). The chain groups $\mathsf{C}_0(\Lambda;\mathbb{A})$, $\mathsf{C}_1(\Lambda;\mathbb{A})$, $\mathsf{C}_2(\Lambda;\mathbb{A})$ are the groups of formal sums $\sum_x a_x\, x$, $\sum_\ell a_\ell\, \ell$, $\sum_p a_p\, p$ with coefficients in $\mathbb{A}$ over sites, links and plaquettes. The boundary maps are the homomorphisms with

$$\partial(x,\lambda) := (x+e_\lambda) - (x), \qquad \partial(x,\lambda\lambda') := (x,\lambda) + (x+e_\lambda,\lambda') - (x+e_{\lambda'},\lambda) - (x,\lambda').$$

For $c = \sum a_\ell \ell$ we write $c_\ell := a_\ell$, and $\operatorname{supp} c$ is the set of cells with non-zero coefficient. The *incidence number* $[p:\ell] \in \{0,\pm 1\}$ is the coefficient of ℓ in ∂p (over $\mathbb{Z}$); it is non-zero iff $\ell \in E(p)$, the four links of p being distinct. By [F, eq. `eq:plaquette`], the plaquette variable is the product, in cyclic order, of the $U_\ell^{[p:\ell]}$, $\ell \in E(p)$. A set S of plaquettes is identified with the chain $\sum_{p\in S} p$. For a closed path $C = (\ell_1^{\epsilon_1}, \dots, \ell_n^{\epsilon_n})$ put $[C] := \sum_i \epsilon_i \ell_i \in \mathsf{C}_1(\Lambda;\mathbb{Z})$ and $[C:\ell] := [C]_\ell$. Reduction modulo M maps $\mathsf{C}_k(\Lambda;\mathbb{Z}) \to \mathsf{C}_k(\Lambda;\mathbb{Z}_M)$ and commutes with ∂. The same definitions, with $d = 2$, apply to any two-dimensional periodic lattice.

Lemma 3.2 (Boundaries)**.** *(i) $\partial\circ\partial = 0$ on $\mathsf{C}_2(\Lambda;\mathbb{A})$. (ii) For a rectangular loop C, $\partial D = [C]$ and $\partial\Sigma = 0$.*

Proof. (i) For $p = (x, \lambda\lambda')$, $\partial\partial p = [(x+e_\lambda) - (x)] + [(x+e_\lambda+e_{\lambda'}) - (x+e_\lambda)] - [(x+e_{\lambda'}+e_\lambda) - (x+e_{\lambda'})] - [(x+e_{\lambda'}) - (x)] = 0$. (ii) Summing the definition of ∂ over D,

$$\begin{aligned}\partial D &= \sum_{i<R}\sum_{j<T}\big[(x_{ij},\mu) - (x_{i,j+1},\mu)\big] + \sum_{j<T}\sum_{i<R}\big[(x_{i+1,j},\nu) - (x_{ij},\nu)\big] \\ &= \sum_{i<R}\big[(x_{i0},\mu) - (x_{iT},\mu)\big] + \sum_{j<T}\big[(x_{Rj},\nu) - (x_{0j},\nu)\big],\end{aligned}$$

by telescoping, and this is $[C]$ by Definition 2.3. For Σ: a link (y,μ) with y in the slice $\{x : x_\lambda = x_\lambda^0 \text{ for } \lambda \notin \{\mu,\nu\}\}$ lies in exactly two plaquettes of Σ, namely $(y,\mu\nu)$ with incidence $+1$ and $(y-e_\nu,\mu\nu)$ with incidence -1 (distinct since $L_\nu \geq 2$); similarly (y,ν) lies in $(y,\mu\nu)$ with incidence -1 and in $(y-e_\mu,\mu\nu)$ with incidence $+1$; all other links lie in no plaquette of Σ. Hence $\partial\Sigma = 0$. □

Lemma 3.3 (Two-cycles of a two-dimensional torus)**.** *Let $L_1, L_2 \geq 2$ and let $\Lambda_2 = \mathbb{Z}/L_1\mathbb{Z} \times \mathbb{Z}/L_2\mathbb{Z}$, with unit vectors e_1, e_2, links $(y,1),(y,2)$ and plaquettes $(y,12)$. For every abelian group $\mathbb{A}$, a chain $c = \sum_y c(y)\,(y,12) \in \mathsf{C}_2(\Lambda_2;\mathbb{A})$ satisfies $\partial c = 0$ if and only if $c(y) = a$ for all y, for some $a \in \mathbb{A}$; i.e. the 2-cycles are the multiples $a\mathbf{1}$ of $\mathbf{1} := \sum_y (y,12)$.*

Proof. Since $\partial(y',12) = (y',1) + (y'+e_1,2) - (y'+e_2,1) - (y',2)$, the link $(y,1)$ occurs exactly in $\partial(y,12)$ (with coefficient $+1$) and in $\partial(y-e_2,12)$ (with coefficient -1), and these plaquettes are distinct because $L_2 \geq 2$. Hence $(\partial c)_{(y,1)} = c(y) - c(y-e_2)$. Likewise $(y,2)$ occurs exactly in $\partial(y,12)$ (coefficient -1) and in $\partial(y-e_1,12)$ (coefficient $+1$), so $(\partial c)_{(y,2)} = c(y-e_1) - c(y)$. Thus $\partial c = 0$ iff c is invariant under the translations by e_1 and e_2, which generate Λ_2, i.e. iff c is constant. □

Definition 3.4 (Projection onto a coordinate 2-torus)**.** Fix $\mu < \nu$ and let $\Lambda_{\mu\nu} := \mathbb{Z}/L_\mu\mathbb{Z} \times \mathbb{Z}/L_\nu\mathbb{Z}$ be the two-dimensional periodic lattice of Lemma 3.3 (with $L_1 = L_\mu$, $L_2 = L_\nu$), $\pi : \Lambda \to \Lambda_{\mu\nu}$, $\pi(x) = (x_\mu, x_\nu)$. Define homomorphisms $\pi_\# : \mathsf{C}_k(\Lambda; \mathbb{A}) \to \mathsf{C}_k(\Lambda_{\mu\nu}; \mathbb{A})$, $k = 1, 2$, on basis elements by

$$\pi_\#(x, \mu) = (\pi x, 1), \quad \pi_\#(x, \nu) = (\pi x, 2), \quad \pi_\#(x, \lambda) = 0 \ \ (\lambda \notin \{\mu, \nu\}),$$

$$\pi_\#(x, \mu\nu) = (\pi x, 12), \qquad \pi_\#(x, \lambda\lambda') = 0 \ \ (\{\lambda, \lambda'\} \neq \{\mu, \nu\}).$$

Lemma 3.5 (The projection is a chain map)**.** *$\partial \circ \pi_\# = \pi_\# \circ \partial$ on $\mathsf{C}_2(\Lambda; \mathbb{A})$.*

Proof. It suffices to check basis elements $p = (x, \lambda\lambda')$, $\lambda < \lambda'$. Note $\pi(x + e_\mu) = \pi x + e_1$, $\pi(x + e_\nu) = \pi x + e_2$ and $\pi(x + e_\tau) = \pi x$ for $\tau \notin \{\mu, \nu\}$.

Case $\{\lambda, \lambda'\} = \{\mu, \nu\}$. Then $\pi_\#\partial p = (\pi x, 1) + (\pi x + e_1, 2) - (\pi x + e_2, 1) - (\pi x, 2) = \partial(\pi x, 12) = \partial\pi_\# p$.

Case $\{\lambda, \lambda'\} \cap \{\mu, \nu\} = \emptyset$. All four links of p are mapped to 0, and $\pi_\# p = 0$.

Case of exactly one common direction. Let $\alpha \in \{\mu, \nu\}$ be the common direction and τ the other direction of p. Then $\pi_\# p = 0$. By the definition of ∂, ∂p consists of the two τ-links of p, which are mapped to 0, and of $\pm[(x, \alpha) - (x + e_\tau, \alpha)]$ (with sign $+$ if $\alpha = \lambda$ and $-$ if $\alpha = \lambda'$). Since $\pi(x + e_\tau) = \pi x$, the two α-links have the same image and $\pi_\#\partial p = 0$. □

Definition 3.6 ($\mathbb{Z}_M$-area)**.** For a closed path C and $m \in \mathbb{Z}_M$ let

$$A_m(C) := \min\{|\operatorname{supp}\zeta| : \ \zeta \in \mathsf{C}_2(\Lambda; \mathbb{Z}_M), \ \partial\zeta = -m[C]\} \in \{0, 1, 2, \dots\} \cup \{+\infty\},$$

with $\min\emptyset := +\infty$ (here $[C]$ is reduced modulo M). Clearly $A_0(C) = 0$.

Proposition 3.7 (Area of a rectangle)**.** *Let C be a rectangular loop in the plane (μ, ν), $m \in \mathbb{Z}_M \setminus \{0\}$, and $\zeta \in \mathsf{C}_2(\Lambda; \mathbb{Z}_M)$ with $\partial\zeta = -m[C]$. Then*

$$\big|\operatorname{supp}\zeta \cap \mathcal{P}_{\mu\nu}\big| \ \geq \ A = \min(RT, \, L_\mu L_\nu - RT).$$

Moreover $A_m(C) = A$.

Proof. By Lemmas 3.5 and 3.2(ii),

$$\partial(\pi_\#\zeta) = \pi_\#(\partial\zeta) = -m\,\pi_\#[C] = -m\,\pi_\#\partial D = -m\,\partial(\pi_\# D).$$

Here $\pi_\# D = D'$ is the set of the RT plaquettes $(\pi x_{ij}, 12)$ of $\Lambda_{\mu\nu}$, which are distinct since $R \leq L_\mu$ and $T \leq L_\nu$. Thus $\pi_\#\zeta + mD'$ is a 2-cycle of $\Lambda_{\mu\nu}$ with coefficients in $\mathbb{Z}_M$, and by Lemma 3.3 there is $k \in \mathbb{Z}_M$ with $\pi_\#\zeta = k\mathbf{1} - mD'$: the coefficient of $\pi_\#\zeta$ is $k - m$ on the RT plaquettes of D' and k on the other $L_\mu L_\nu - RT$ plaquettes. If $k = 0$ it is non-zero on D' (as $m \neq 0$); if $k = m$ it is non-zero off D'; otherwise it is non-zero everywhere. In all cases the number of plaquettes q of $\Lambda_{\mu\nu}$ with $(\pi_\#\zeta)_q \neq 0$ is at least A. Now $(\pi_\#\zeta)_q = \sum_{p \in \pi^{-1}(q)} \zeta_p$, where $\pi^{-1}(q) := \{(x, \mu\nu) : \pi x = q\} \subseteq \mathcal{P}_{\mu\nu}$; so $(\pi_\#\zeta)_q \neq 0$ implies $\operatorname{supp}\zeta \cap \pi^{-1}(q) \neq \emptyset$, and the sets $\pi^{-1}(q)$ are pairwise disjoint. Hence $|\operatorname{supp}\zeta \cap \mathcal{P}_{\mu\nu}| \geq A$, and in particular $A_m(C) \geq A$. Conversely, by Lemma 3.2(ii) the chains $-mD$ and $m(\Sigma - D)$ have boundary $-m[C]$ and supports D and $\Sigma \setminus D$, of sizes RT and $L_\mu L_\nu - RT$; so $A_m(C) \leq A$. □

4 The centre selection rule

Throughout this section $z_0 \in G$ is central of finite order $M \geq 2$, $\omega = e^{2\pi i/M}$, and $m(r)$ is the z_0-charge (Definition 2.5).

Lemma 4.1 (Charge decomposition of f_β)**.** *For $k \in \mathbb{Z}_M$ and $U \in G$ define*

$$f_k(U) := \frac{1}{M} \sum_{j=0}^{M-1} \omega^{-jk} f_\beta(z_0^j U).$$

Then:

(i) f_k is continuous and $f_k(z_0^j U) = \omega^{jk} f_k(U)$ for all $j \in \mathbb{Z}$, $U \in G$;

(ii) $\sum_{k \in \mathbb{Z}_M} f_k = f_\beta$;

(iii) $\|f_k\|_\infty \le \epsilon(\beta)$;

(iv) $f_k = \sum_{r \neq 0,\ m(r)=k} \frac{c_r(\beta)}{c_0(\beta)} \chi_r$, the series converging absolutely and uniformly.

Only (i) and (ii) are used below.

Proof. (i) Continuity is clear. Since $z_0^M = 1$, $j \mapsto \omega^{-jk} f_\beta(z_0^j U)$ is M-periodic, so shifting the summation index, $f_k(z_0 U) = \frac{1}{M} \sum_j \omega^{-jk} f_\beta(z_0^{j+1} U) = \omega^k f_k(U)$; iterating gives the claim for $j \ge 0$, and for $j < 0$ use $z_0^{-1} = z_0^{M-1}$. (ii) $\sum_k f_k(U) = \frac{1}{M} \sum_j f_\beta(z_0^j U) \sum_k \omega^{-jk} = f_\beta(U)$, because $\sum_{k \in \mathbb{Z}_M} \omega^{-jk} = M \delta_{j0}$ for $0 \le j < M$. (iii) is the triangle inequality and (3). (iv) By [F, Corollary `cor:uniform`], $f_\beta = \sum_{r \neq 0} (c_r/c_0) \chi_r$ absolutely and uniformly. Inserting this into the (finite) sum defining f_k and using Lemma 2.4, $f_k = \sum_{r \neq 0} (c_r/c_0) \chi_r \cdot \frac{1}{M} \sum_j \omega^{j(m(r)-k)} = \sum_{r \neq 0,\, m(r)=k} (c_r/c_0) \chi_r$. □

Definition 4.2 (Loop integrals)**.** For a closed path C, an irreducible representation r, a set $X \subseteq \mathcal{P}$ and a map $\zeta : X \to \mathbb{Z}_M$, put

$$\Phi_{r,C}(X) := \int \chi_r(U_C) \prod_{p \in X} f_\beta(U_p)\, dU, \qquad \Phi_{r,C}(X;\zeta) := \int \chi_r(U_C) \prod_{p \in X} f_{\zeta(p)}(U_p)\, dU,$$

and identify ζ with the chain $\sum_{p \in X} \zeta(p)\, p \in \mathsf{C}_2(\Lambda; \mathbb{Z}_M)$, whose support is contained in X.

Lemma 4.3 (Selection rule)**.** *If $\Phi_{r,C}(X;\zeta) \neq 0$, then $\partial \zeta = -m(r)\,[C]$ in $\mathsf{C}_1(\Lambda; \mathbb{Z}_M)$.*

Proof. Fix a link ℓ and let $\tau_\ell : \Omega_\Lambda \to \Omega_\Lambda$ replace U_ℓ by $z_0 U_\ell$ and leave the other coordinates unchanged. It preserves dU (left invariance of Haar measure in the ℓ-th factor). Since z_0 is central and ℓ occurs at most once among the links of a plaquette, $(\tau_\ell U)_p = z_0^{[p:\ell]} U_p$ for every plaquette p (Definition 3.1); and $(\tau_\ell U)_C = z_0^{[C:\ell]} U_C$, since every occurrence of ℓ in C contributes a factor $z_0^{\pm 1}$ which can be moved to the front. Let $F(U) := \chi_r(U_C) \prod_{p \in X} f_{\zeta(p)}(U_p)$. By Lemma 4.1(i) and Lemma 2.4,

$$F(\tau_\ell U) = \omega^{\sum_{p \in X} [p:\ell] \zeta(p) + m(r)[C:\ell]}\, F(U) = \omega^{(\partial \zeta + m(r)[C])_\ell}\, F(U),$$

the exponent being computed in $\mathbb{Z}_M$. Integrating, $\Phi_{r,C}(X;\zeta) = \omega^{(\partial\zeta + m(r)[C])_\ell}\, \Phi_{r,C}(X;\zeta)$. If $\Phi_{r,C}(X;\zeta) \neq 0$, then $\omega^{(\partial\zeta + m(r)[C])_\ell} = 1$, i.e. $(\partial\zeta + m(r)[C])_\ell = 0$ in $\mathbb{Z}_M$, because ω is a primitive M-th root of unity. This holds for every ℓ. □

Proposition 4.4 (Vanishing of small loop polymers)**.** *Let C be a closed path, r irreducible and $X \subseteq \mathcal{P}$. If $\Phi_{r,C}(X) \neq 0$, then there is $\zeta \in \mathsf{C}_2(\Lambda; \mathbb{Z}_M)$ with $\operatorname{supp} \zeta \subseteq X$ and $\partial \zeta = -m(r)[C]$; in particular $|X| \ge A_{m(r)}(C)$. If C is a rectangular loop in the plane (μ, ν) and $m(r) \neq 0$, then $\Phi_{r,C}(X) \neq 0$ implies $|X \cap \mathcal{P}_{\mu\nu}| \ge A$.*

Proof. By Lemma 4.1(ii), $\prod_{p\in X} f_\beta(U_p) = \sum_{\zeta\in\mathbb{Z}_M^X} \prod_{p\in X} f_{\zeta(p)}(U_p)$, so $\Phi_{r,C}(X) = \sum_\zeta \Phi_{r,C}(X;\zeta)$ (a finite sum). If $\Phi_{r,C}(X) \neq 0$, some $\Phi_{r,C}(X;\zeta) \neq 0$, and Lemma 4.3 applies to this ζ. Then $|X| \geq |\operatorname{supp}\zeta| \geq A_{m(r)}(C)$, and for a rectangle Proposition 3.7 gives $|X \cap \mathcal{P}_{\mu\nu}| \geq |\operatorname{supp}\zeta \cap \mathcal{P}_{\mu\nu}| \geq A$. □

Proposition 4.5 (Loops winding around the torus)**.** *Let C be a closed path, λ a direction, and $w_\lambda(C) := \sum_{x:\,x_\lambda=0}[C : (x,\lambda)] \in \mathbb{Z}$ the net number of times C crosses from the slice $x_\lambda = 0$ to the slice $x_\lambda = 1$. If $m(r)\, w_\lambda(C) \neq 0$ in $\mathbb{Z}_M$, then $A_{m(r)}(C) = +\infty$, and $\langle\chi_r(U_C)\rangle_{\Lambda,\beta} = 0$ for every $\beta \geq 0$.*

Proof. Let $\phi_\lambda : \mathsf{C}_1(\Lambda;\mathbb{Z}_M) \to \mathbb{Z}_M$, $\phi_\lambda(c) := \sum_{x:\,x_\lambda=0} c_{(x,\lambda)}$. A plaquette not spanning direction λ has no λ-links; a plaquette spanning λ and λ' has exactly two λ-links, (y,λ) and $(y + e_{\lambda'},\lambda)$, with opposite incidence numbers and the same λ-coordinate. Hence $\phi_\lambda(\partial p) = 0$ for every plaquette, so $\phi_\lambda \circ \partial = 0$. If $\partial\zeta = -m(r)[C]$, then $0 = \phi_\lambda(\partial\zeta) = -m(r)w_\lambda(C)$ in $\mathbb{Z}_M$, a contradiction; so $A_{m(r)}(C) = +\infty$. For the second claim let τ replace U_ℓ by $z_0 U_\ell$ simultaneously for all links $\ell = (x,\lambda)$ with $x_\lambda = 0$. It preserves dU. By the observation just made, a plaquette contains either no transformed link or exactly two, ℓ and ℓ', with $[p:\ell]+[p:\ell'] = 0$; as z_0 is central, $(\tau U)_p = z_0^{[p:\ell]+[p:\ell']} U_p = U_p$ in the second case, so $W_\beta \circ \tau = W_\beta$. Moreover $(\tau U)_C = z_0^{w_\lambda(C)} U_C$. Hence $\langle\chi_r(U_C)\rangle = \langle\chi_r(U_C)\circ\tau\rangle = \omega^{m(r)w_\lambda(C)}\langle\chi_r(U_C)\rangle$, and the expectation vanishes. □

Remark 4.6 (Why the charge decomposition is used only inside Φ)*.* One could decompose every factor $f_\beta(U_p)$ of $\prod_p(1+f_\beta(U_p))$ as $\sum_k f_k(U_p)$ from the start, i.e. expand over assignments $\mathcal{P} \to \{*\} \cup \mathbb{Z}_M$. The selection rule would then apply term by term, but bounding each term separately costs a factor $M^{|X|}$ (or requires the character coefficients). Proposition 4.4 uses the decomposition only to show that $\Phi_{r,C}(X)$ *vanishes*; the size of the non-vanishing $\Phi_{r,C}(X)$ is then estimated with the undecomposed f_β by $d_r\epsilon(\beta)^{|X|}$. This is why no constant below depends on M or N.

5 Expansion of a Wilson loop and the ratio of partition functions

Definition 5.1 (Restricted partition functions, loop polymers)**.** For $\mathcal{Q} \subseteq \mathcal{P}$ let

$$\Xi(\mathcal{Q}) := \int \prod_{p\in\mathcal{Q}} (1 + f_\beta(U_p))\, dU = c_0(\beta)^{-|\mathcal{Q}|} \int \prod_{p\in\mathcal{Q}} w_\beta(U_p)\, dU \ \in (0,\infty),$$

and let $\Gamma(\mathcal{Q})$ be the set of polymers contained in $\mathcal{Q}$. Thus $\Xi(\mathcal{P}) = Z_\Lambda(\beta)/c_0(\beta)^{|\mathcal{P}|}$ is the polymer partition function $\Xi_\Lambda(\beta)$ of [F, Proposition 'Polymer representation' (`prop:polymer`)]. For a closed path C let $\mathcal{P}_C := \{p \in \mathcal{P} : E(p) \cap E_C \neq \emptyset\}$; let $\mathcal{X}_C$ be the family of sets $X \subseteq \mathcal{P}$ every connected component of which contains a plaquette of $\mathcal{P}_C$ (including $X = \emptyset$); and for $X \subseteq \mathcal{P}$ let

$$\overline{X} := \{p \in \mathcal{P} :\ E(p) \cap (E(X) \cup E_C) \neq \emptyset\} \ \supseteq\ X \cup \mathcal{P}_C.$$

By Lemma 2.2(d), $|\mathcal{P}_C| \leq 2(d-1)|E_C|$ and $|\overline{X}| \leq (K+1)|X| + 2(d-1)|E_C|$.

For an abstract polymer system we use square brackets: if Γ is a finite set with a symmetric and reflexive relation $\not\approx$ (incompatibility) and $z : \Gamma \to \mathbb{C}$, then for $\mathcal{F} \subseteq \Gamma$

$$\Xi[\mathcal{F}] := \sum_{\substack{\mathcal{S}\subseteq\mathcal{F} \\ \text{pairwise compatible}}} \prod_{\gamma\in\mathcal{S}} z(\gamma), \tag{4}$$

the empty family contributing 1. Since $\gamma \not\approx \gamma$, pairwise compatible tuples have distinct entries, so this agrees with the sum over ordered tuples weighted by $1/n!$ used in [F].

Lemma 5.2 (Polymer representation of $\Xi(\mathcal{Q})$)**.** *For every $\mathcal{Q} \subseteq \mathcal{P}$, $\Xi(\mathcal{Q}) = \Xi[\Gamma(\mathcal{Q})]$, for the polymer system of [F, Definition* `def:polymer`*] restricted to $\Gamma(\mathcal{Q})$.*

Proof. Expanding the product, $\Xi(\mathcal{Q}) = \sum_{Y \subseteq \mathcal{Q}} \int \prod_{p \in Y} f_\beta(U_p)\, dU$. The connected components $\gamma_1, \dots, \gamma_k$ of Y are polymers in $\Gamma(\mathcal{Q})$, pairwise compatible (two components sharing a link would be adjacent), and the integral factorises into $\prod_i z(\gamma_i)$ because the γ_i depend on disjoint sets of link variables and dU is a product measure. Conversely, a pairwise compatible family in $\Gamma(\mathcal{Q})$ consists of disjoint polymers no two of which contain adjacent plaquettes, so it is the family of components of its union. Hence $Y \mapsto \{\gamma_1, \dots, \gamma_k\}$ is a bijection onto the compatible subfamilies of $\Gamma(\mathcal{Q})$ (this is the argument of [F, Proposition `prop:polymer`]). Positivity of $\Xi(\mathcal{Q})$ follows from $1 + f_\beta = w_\beta / c_0 > 0$. □

Lemma 5.3 (Loop expansion)**.** *For every closed path C, irreducible r and $\beta \geq 0$,*

$$\langle \chi_r(U_C) \rangle_{\Lambda,\beta} = \sum_{X \in \mathcal{X}_C} \Phi_{r,C}(X)\, \frac{\Xi(\mathcal{P} \setminus \overline{X})}{\Xi(\mathcal{P})}.$$

Proof. Since $w_\beta = c_0(1 + f_\beta)$,

$$\frac{Z_\Lambda(\beta)}{c_0^{|\mathcal{P}|}} \langle \chi_r(U_C) \rangle = \int \chi_r(U_C) \prod_{p \in \mathcal{P}} (1 + f_\beta(U_p))\, dU = \sum_{S \subseteq \mathcal{P}} \int \chi_r(U_C) \prod_{p \in S} f_\beta(U_p)\, dU.$$

For $S \subseteq \mathcal{P}$ let $X(S)$ be the union of the components of S that meet $\mathcal{P}_C$, and $Y(S) := S \setminus X(S)$.

Claim: $S \mapsto (X(S), Y(S))$ is a bijection from the subsets of $\mathcal{P}$ onto the pairs (X, Y) with $X \in \mathcal{X}_C$ and $Y \subseteq \mathcal{P} \setminus \overline{X}$, with inverse $(X, Y) \mapsto X \cup Y$. Clearly $X(S) \in \mathcal{X}_C$. Let $p \in Y(S)$. Its component in S does not meet $\mathcal{P}_C$, so $E(p) \cap E_C = \emptyset$; and $E(p) \cap E(X(S)) = \emptyset$, since otherwise p would share a link with a plaquette of $X(S)$ and belong to one of the components forming $X(S)$. So $Y(S) \cap \overline{X(S)} = \emptyset$. Conversely let $X \in \mathcal{X}_C$, $Y \subseteq \mathcal{P} \setminus \overline{X}$ and $S = X \cup Y$ (a disjoint union, as $X \subseteq \overline{X}$). No plaquette of Y shares a link with a plaquette of X, so every component of S is a component of X or a component of Y; the former meet $\mathcal{P}_C$, the latter do not (as $\mathcal{P}_C \subseteq \overline{X}$). Hence $X(S) = X$ and $Y(S) = Y$, which proves the claim.

For such a pair, the function $\chi_r(U_C) \prod_{p \in X} f_\beta(U_p)$ depends only on the links in $E_C \cup E(X)$, and $\prod_{p \in Y} f_\beta(U_p)$ only on the links in $E(Y)$, which is disjoint from $E_C \cup E(X)$ because $Y \cap \overline{X} = \emptyset$. As dU is a product measure, the integral factorises, and

$$\begin{aligned} \sum_{S \subseteq \mathcal{P}} \int \chi_r(U_C) \prod_{p \in S} f_\beta(U_p)\, dU &= \sum_{X \in \mathcal{X}_C} \Phi_{r,C}(X) \sum_{Y \subseteq \mathcal{P} \setminus \overline{X}} \int \prod_{p \in Y} f_\beta(U_p)\, dU \\ &= \sum_{X \in \mathcal{X}_C} \Phi_{r,C}(X)\, \Xi(\mathcal{P} \setminus \overline{X}). \end{aligned}$$

Dividing by $Z_\Lambda / c_0^{|\mathcal{P}|} = \Xi(\mathcal{P})$ gives the lemma. □

The following lemma is the inductive "no-cluster-expansion" form of the Kotecký–Preiss criterion [KP86, Dob96]; see [FP07, Section 1] for its history. The same bound also follows from [FP07, (2.11), (2.14), (2.15)], where it is derived from the cluster (tree-graph) expansion. We give the short inductive proof.

Lemma 5.4 (Ratio bound for polymer systems)**.** *Let Γ be a finite set with a symmetric reflexive relation $\not\approx$, let $z : \Gamma \to \mathbb{C}$, and suppose $a : \Gamma \to [0, \infty)$ satisfies*

$$\sum_{\gamma' \in \Gamma:\ \gamma' \not\approx \gamma} |z(\gamma')|\, e^{a(\gamma')} \leq a(\gamma) \qquad \text{for all } \gamma \in \Gamma. \tag{5}$$

Put $u(\gamma) := |z(\gamma)|e^{a(\gamma)}$. Then for every $\mathcal{F} \subseteq \Gamma$, $\Xi[\mathcal{F}] \neq 0$, and for every $\gamma \in \mathcal{F}$

$$e^{-u(\gamma)} \leq \Big|\frac{\Xi[\mathcal{F}]}{\Xi[\mathcal{F} \setminus \{\gamma\}]}\Big| \leq e^{u(\gamma)}. \tag{6}$$

Consequently, for $\mathcal{F}' \subseteq \mathcal{F} \subseteq \Gamma$,

$$\exp\Big(-\sum_{\gamma \in \mathcal{F}\setminus\mathcal{F}'} u(\gamma)\Big) \leq \Big|\frac{\Xi[\mathcal{F}']}{\Xi[\mathcal{F}]}\Big| \leq \exp\Big(\sum_{\gamma \in \mathcal{F}\setminus\mathcal{F}'} u(\gamma)\Big).$$

Proof. We prove by induction on $n \geq 0$: *for every $\mathcal{F}$ with $|\mathcal{F}| \leq n$, $\Xi[\mathcal{F}] \neq 0$ and (6) holds for every $\gamma \in \mathcal{F}$.* For $n = 0$, $\Xi[\emptyset] = 1$. Let $|\mathcal{F}| = n+1$ and $\gamma \in \mathcal{F}$, and let $\mathcal{N}(\gamma) := \{\gamma' : \gamma' \not\approx \gamma\} \ni \gamma$. A compatible family $\mathcal{S} \subseteq \mathcal{F}$ either does not contain γ, or contains γ and then $\mathcal{S} \setminus \{\gamma\}$ is a compatible family in $\mathcal{F} \setminus \mathcal{N}(\gamma)$; conversely every such family together with γ is compatible. Hence, with $\mathcal{F}_0 := \mathcal{F} \setminus \{\gamma\}$,

$$\Xi[\mathcal{F}] = \Xi[\mathcal{F}_0] + z(\gamma)\,\Xi[\mathcal{F} \setminus \mathcal{N}(\gamma)]. \tag{7}$$

Let $\gamma_1, \dots, \gamma_k$ be the elements of $\mathcal{F}_0 \cap \mathcal{N}(\gamma)$ and $\mathcal{F}_j := \mathcal{F}_{j-1} \setminus \{\gamma_j\}$, so that $\mathcal{F}_k = \mathcal{F} \setminus \mathcal{N}(\gamma)$. Each $\mathcal{F}_{j-1}$ has at most n elements and contains γ_j, so by the induction hypothesis $\Xi[\mathcal{F}_j] \neq 0$ for $0 \leq j \leq k$ and $|\Xi[\mathcal{F}_j]| \leq e^{u(\gamma_j)}|\Xi[\mathcal{F}_{j-1}]|$ (left inequality in (6)). Therefore, using (5) at γ (the sum there contains γ itself and the distinct γ_j),

$$|\Xi[\mathcal{F} \setminus \mathcal{N}(\gamma)]| \leq e^{\sum_j u(\gamma_j)}\,|\Xi[\mathcal{F}_0]| \leq e^{a(\gamma)-u(\gamma)}\,|\Xi[\mathcal{F}_0]|.$$

By (7), $\Xi[\mathcal{F}] = \Xi[\mathcal{F}_0](1+w)$ with $|w| \leq |z(\gamma)|e^{a(\gamma)-u(\gamma)} = u(\gamma)e^{-u(\gamma)}$. Writing $u = u(\gamma)$, the elementary inequality $e^u \geq 1+u$ gives $1 - ue^{-u} \geq e^{-u}$ and $1 + ue^{-u} \leq 1 + u \leq e^u$, so $e^{-u} \leq |1+w| \leq e^u$. This is (6), and it shows $\Xi[\mathcal{F}] \neq 0$. The last statement follows by removing the elements of $\mathcal{F} \setminus \mathcal{F}'$ one at a time and applying (6) at each step. □

Lemma 5.5 (The Kotecký–Preiss condition for plaquette polymers)**.** *Let $0 < \beta < \beta_0(d)$. For every set $S \subseteq \mathcal{P}$,*

$$\sum_{\gamma \text{ polymer: } \gamma \cap S \neq \emptyset} |z(\gamma)|\, e^{|\gamma|} \;\leq\; \theta(\beta)\,|S|.$$

In particular, for every polymer γ_0, $\sum_{\gamma \not\approx \gamma_0} |z(\gamma)|e^{|\gamma|} \leq (K+1)\theta(\beta)|\gamma_0| \leq |\gamma_0|$, i.e. (5) holds for the polymers of Λ with $a(\gamma) = |\gamma|$.

Proof. By Lemma 2.2(e),(f),

$$\sum_{\gamma \cap S \neq \emptyset} |z(\gamma)|e^{|\gamma|} \leq \sum_{p \in S} \sum_{n \geq 1} \sum_{\gamma \ni p,\, |\gamma| = n} (e\epsilon)^n \leq |S| \sum_{n \geq 1} (e^2 K \epsilon)^n = |S|\,\theta.$$

A polymer $\gamma \not\approx \gamma_0$ contains a plaquette sharing a link with a plaquette of γ_0, i.e. meets the set S of plaquettes equal or adjacent to a plaquette of γ_0, and $|S| \leq (K+1)|\gamma_0|$; finally $(K+1)\theta < 1$ by Lemma 2.8. (This is the computation in the proof of [F, Theorem 'Convergence of the strong-coupling cluster expansion' (`thm:conv`)].) □

Proposition 5.6 (Removing plaquettes)**.** *Let $0 < \beta < \beta_0(d)$. For every $\mathcal{Q} \subseteq \mathcal{P}$,*

$$e^{-\theta(\beta)|\mathcal{P}\setminus\mathcal{Q}|} \;\leq\; \frac{\Xi(\mathcal{Q})}{\Xi(\mathcal{P})} \;\leq\; e^{\theta(\beta)|\mathcal{P}\setminus\mathcal{Q}|}.$$

Proof. By Lemma 5.2, $\Xi(\mathcal{Q}) = \Xi[\Gamma(\mathcal{Q})]$ and $\Xi(\mathcal{P}) = \Xi[\Gamma(\mathcal{P})]$, and both are positive. By Lemma 5.5, Lemma 5.4 applies to $\Gamma(\mathcal{P})$ with $a(\gamma) = |\gamma|$. The polymers in $\Gamma(\mathcal{P}) \setminus \Gamma(\mathcal{Q})$ are exactly those meeting $S := \mathcal{P} \setminus \mathcal{Q}$, and for them $\sum u(\gamma) \leq \theta|S|$ by Lemma 5.5. □

Remark 5.7. One can also obtain Proposition 5.6, with $e^{\pm|\mathcal{P}\setminus\mathcal{Q}|}$ instead of $e^{\pm\theta|\mathcal{P}\setminus\mathcal{Q}|}$, from the cluster expansion: by [F, Cited Theorem 'Kotecký–Preiss 1986' (`thm:KP`)] applied to $\Gamma(\mathcal{P})$ and to $\Gamma(\mathcal{Q})$, $\log\Xi(\mathcal{P}) - \log\Xi(\mathcal{Q})$ is the sum of the cluster weights over the clusters containing a polymer which meets $\mathcal{P}\setminus\mathcal{Q}$, and by the last statement of [F, Theorem `thm:conv`] these weigh at most 1 per plaquette of $\mathcal{P}\setminus\mathcal{Q}$. We have preferred the self-contained Lemma 5.4.

Lemma 5.8 (Rooted counting)**.** *Let $\mathcal{G}$ be a simple graph on a finite vertex set V, $o \in V$, and $J, K \geq 1$ integers such that o has at most J neighbours and every $v \in V\setminus\{o\}$ has at most K neighbours in $V\setminus\{o\}$. For $n \geq 0$, the number of sets $X \subseteq V\setminus\{o\}$ with $|X| = n$ such that the subgraph induced on $X\cup\{o\}$ is connected is at most $\binom{J+Kn}{n}$, hence at most $(1+t)^J((1+t)^K/t)^n$ for every $t > 0$.*

Proof. Label the neighbours of o injectively by elements of $\{1,\dots,J\}$ and, for each $v \neq o$, the neighbours of v in $V\setminus\{o\}$ injectively by elements of $\{1,\dots,K\}$. Given X, let H be the subgraph induced on $X\cup\{o\}$ and run depth-first search on H from o: from the current vertex move to its unvisited H-neighbour of smallest label, and go back to the parent when there is none. As H is connected, this visits all of $X\cup\{o\}$ and produces a spanning tree τ of H rooted at o. The children of o are neighbours of o; since o is visited first, the children of $v \neq o$ are neighbours of v in $V\setminus\{o\}$. The children of a vertex are visited in increasing order of labels (a child chosen later was unvisited, hence available, when an earlier child was chosen), and the subtree of a child is visited completely before the next child. List the vertices in the order of their first visit, $o = v_0, v_1, \dots, v_n$, and let $b_0 \in \{0,1\}^J$ and $b_i \in \{0,1\}^K$ $(i \geq 1)$ be the words whose j-th letter is 1 iff the neighbour of v_i with label j is a child of v_i in τ. The word $b = b_0 b_1 \cdots b_n$ has length $J + Kn$ and exactly n letters 1 (one per non-root vertex).

The word b determines X. Indeed, by induction on i, the blocks $b_0,\dots,b_i$ determine $v_0,\dots,v_i$ and their children: $v_0 = o$, and v_{i+1} is the not yet visited child of smallest label of the most recently visited vertex among $v_0,\dots,v_i$ that still has a not yet visited child (at every time, the vertices with unvisited children are ancestors of the current vertex, and depth-first search continues from the deepest of them); the children of v_{i+1} are then read off from b_{i+1} and the labelling at v_{i+1}. Hence $X = \{v_1,\dots,v_n\}$ is a function of b, and the number of sets X is at most the number of such words, $\binom{J+Kn}{n}$. Finally $\binom{m}{n}t^n \leq (1+t)^m$ by the binomial theorem. □

Corollary 5.9 (Counting loop polymers)**.** *For every closed path C and $n \geq 0$, $\#\{X \in \mathcal{X}_C : |X| = n\} \leq \binom{J_C+Kn}{n}$ with $J_C := 2(d-1)|E_C|$.*

Proof. Let $\mathcal{G}$ be the graph on $\mathcal{P}\cup\{o\}$ (o a new vertex) whose edges are the pairs of adjacent plaquettes and the pairs $\{o,p\}$ with $p \in \mathcal{P}_C$. Then o has $|\mathcal{P}_C| \leq J_C$ neighbours and each plaquette at most K neighbours in $\mathcal{P}$ (Lemma 2.2(d)). A set $X \subseteq \mathcal{P}$ belongs to $\mathcal{X}_C$ iff $X\cup\{o\}$ induces a connected subgraph: if every component of X contains a plaquette of $\mathcal{P}_C$, each component is joined to o; conversely, if $X\cup\{o\}$ is connected and Y is a component of X, a shortest path from o to Y in $X\cup\{o\}$ leaves o through some $p \in \mathcal{P}_C \cap X$ and then stays in X, hence in the component of p, so $p \in Y$. Apply Lemma 5.8. □

6 The area law

Proposition 6.1 (Master estimate)**.** *Let $0 < \beta < \beta_0(d)$, let C be a closed path and r an irreducible representation, and let $n_0 \geq 0$ be an integer such that $\Phi_{r,C}(X) = 0$ for all $X \in \mathcal{X}_C$ with $|X| < n_0$. Then for every $t > 0$ with $q_t := \frac{(1+t)^K}{t}\, e\,\epsilon(\beta) < 1$,*

$$\big|\langle\chi_r(U_C)\rangle_{\Lambda,\beta}\big| \leq \frac{d_r}{1-q_t}\, q_t^{n_0}\, \exp\Big(2(d-1)\big[\log(1+t) + \theta(\beta)\big]\,|E_C|\Big).$$

Proof. By Lemma 5.3 and the hypothesis,

$$|\langle\chi_r(U_C)\rangle| \le \sum_{X\in\mathcal{X}_C,\,|X|\ge n_0} |\Phi_{r,C}(X)|\,\frac{\Xi(\mathcal{P}\setminus\overline{X})}{\Xi(\mathcal{P})}.$$

For each X, $|\Phi_{r,C}(X)| \le d_r\epsilon^{|X|}$ by $|\chi_r| \le d_r$ and (3), and by Proposition 5.6 and Definition 5.1, $\Xi(\mathcal{P}\setminus\overline{X})/\Xi(\mathcal{P}) \le e^{\theta|\overline{X}|} \le e^{\theta(K+1)|X|+\theta J_C}$, where $J_C := 2(d-1)|E_C|$. Since $(K+1)\theta < 1$ (Lemma 2.8), Corollary 5.9 and Lemma 5.8 give

$$|\langle\chi_r(U_C)\rangle| \le d_r e^{\theta J_C}\sum_{n\ge n_0}\binom{J_C+Kn}{n}(e\epsilon)^n \le d_r e^{\theta J_C}(1+t)^{J_C}\sum_{n\ge n_0} q_t^n = \frac{d_r\,q_t^{n_0}}{1-q_t}\,(1+t)^{J_C}e^{\theta J_C}.$$

□

Theorem 6.2 (Area law, general form)**.** *Let G,ρ be as in [F, Convention 'Group'], let $d \ge 2$, all $L_\lambda \ge 3$, and $0 < \beta < \beta_0(d)$. Let $z_0 \in G$ be central of finite order $M \ge 2$, r an irreducible representation with z_0-charge $m = m(r)$, and C a closed path. If $A_m(C) = +\infty$ then $\langle\chi_r(U_C)\rangle_{\Lambda,\beta} = 0$; otherwise*

$$\big|\langle\chi_r(U_C)\rangle_{\Lambda,\beta}\big| \le \frac{d_r}{1-x(\beta)}\,x(\beta)^{A_m(C)}\,e^{\kappa_1(\beta)|E_C|} = \frac{d_r}{1-x(\beta)}\exp(-\sigma(\beta)A_m(C)+\kappa_1(\beta)|E_C|).$$

Proof. If $A_m(C) = +\infty$, Proposition 4.4 gives $\Phi_{r,C}(X) = 0$ for all X, and the expectation vanishes by Lemma 5.3. Otherwise Proposition 4.4 shows that Proposition 6.1 applies with $n_0 = A_m(C)$. Take $t = 1/K$: then $(1+t)^K \le e$, so $q_t \le e^2K\epsilon = x < 1$, and the claim follows because $q_t \mapsto q_t^{n_0}/(1-q_t)$ is increasing on $[0,1)$ and $2(d-1)[\log(1+\frac{1}{K})+\theta] = \kappa_1$. □

Corollary 6.3 (Area law for rectangular loops)**.** *In the setting of Theorem 6.2, let C be a rectangular loop with sides R,T in the plane (μ,ν) and $m(r) \ne 0$. Then, with $A = \min(RT, L_\mu L_\nu - RT)$ and $|C| = 2(R+T)$,*

$$\big|\langle\chi_r(U_C)\rangle_{\Lambda,\beta}\big| \le d_r\exp(-\sigma(\beta)A+\kappa(\beta)|C|).$$

Proof. $A_m(C) = A$ by Proposition 3.7 and $|E_C| = |C| \ge 4$, so $\frac{1}{1-x} \le \exp(\frac{|C|}{4}\log\frac{1}{1-x})$; insert into Theorem 6.2. □

Corollary 6.4 ($SU(N)$; Theorem 1.1)**.** *Theorem 1.1 holds.*

Proof. Apply Corollary 6.3 with $z_0 = e^{2\pi i/N}\mathbb{1}_N$, $M = N$, $m(r) = n(r)$, and Lemma 2.8 for the properties of σ and κ. □

Corollary 6.5 ($U(1)$ and $U(N)$)**.** *Let $0 < \beta < \beta_0(d)$ and let C be a rectangular loop as in Definition 2.3.*

(i) $G = U(1)$, $\rho = \mathrm{id}$ (so $N = 1$ and $w_\beta(e^{i\phi}) = e^{\beta\cos\phi}$): for every $q \in \mathbb{Z}\setminus\{0\}$, $|\langle U_C^q\rangle_{\Lambda,\beta}| \le \exp(-\sigma(\beta)A+\kappa(\beta)|C|)$.

(ii) $G = U(N)$, ρ the defining representation: if r is irreducible and $D^r(e^{i\phi}\mathbb{1}) = e^{iq\phi}\mathbb{1}$ with $q \ne 0$, then $|\langle\chi_r(U_C)\rangle_{\Lambda,\beta}| \le d_r\exp(-\sigma(\beta)A+\kappa(\beta)|C|)$.

Proof. (ii) Let $M := |q|+1 \ge 2$ and $z_0 := e^{2\pi i/M}\mathbb{1}$, a central element of order M. Then $D^r(z_0) = e^{2\pi iq/M}\mathbb{1}$, so $m(r) = q \bmod M \ne 0$ because $0 < |q| < M$; apply Corollary 6.3. (Every irreducible r of $U(N)$ is of this form for some $q \in \mathbb{Z}$: by Schur's lemma, as in the proof of Lemma 2.4, D^r restricted to the centre $\{e^{i\phi}\mathbb{1}\}$ is a continuous homomorphism $U(1)\to U(1)$, i.e. $e^{i\phi}\mapsto e^{iq\phi}$.) (i) is the case $N = 1$, $\chi_q(U) = U^q$, $d_r = 1$. □

Corollary 6.6 (Perimeter coefficient tending to zero)**.** *In the setting of Corollary 6.3, put* $q(\beta) := e^{1+K\sqrt{\epsilon(\beta)}}\sqrt{\epsilon(\beta)}$*. Then* $q(\beta) < \frac{3}{10}$ *and*

$$\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \le \frac{d_r}{1-q(\beta)}\, q(\beta)^A \exp\Big(2(d-1)\big[\sqrt{\epsilon(\beta)}+\theta(\beta)\big]|C|\Big).$$

As $\beta \downarrow 0$*, the perimeter coefficient* $2(d-1)[\sqrt{\epsilon}+\theta]$ *tends to* 0*,* $\frac{1}{1-q(\beta)} \to 1$*, and* $\log\frac{1}{q(\beta)} = \frac{1}{2}\log\frac{1}{\epsilon(\beta)} - 1 - K\sqrt{\epsilon(\beta)} \to \infty$*.*

Proof. Apply Proposition 6.1 with $n_0 = A$ (Propositions 4.4, 3.7) and $t = \sqrt{\epsilon}$: $q_t = (1+\sqrt{\epsilon})^K e\sqrt{\epsilon} \le q(\beta)$ and $\log(1+t) \le \sqrt{\epsilon}$. Since $\epsilon < \frac{1}{e^2K(K+2)}$, $\sqrt{\epsilon} < \frac{1}{e\sqrt{K(K+2)}} < \frac{1}{eK}$, so $K\sqrt{\epsilon} < \frac{1}{e}$ and $q(\beta) < e^{1+1/e}\cdot\frac{1}{e\sqrt{K(K+2)}} = \frac{e^{1/e}}{\sqrt{K(K+2)}} \le \frac{e^{1/e}}{\sqrt{24}} < 0.295$, using $K \ge 4$. □

7 Infinite volume and the string tension

Definition 7.1 (Periodisation and infinite-volume limit points)**.** Let $\Omega_\infty := G^{E(\mathbb{Z}^d)}$ with the product topology (a compact metrisable space when G is a compact Lie group). For a periodic lattice Λ let $\iota_\Lambda : \Omega_\Lambda \to \Omega_\infty$, $(\iota_\Lambda U)_{(x,\lambda)} := U_{([x],\lambda)}$, where $[x]$ is the image of $x \in \mathbb{Z}^d$ in Λ, and let $\tilde\mu_{\Lambda,\beta} := (\iota_\Lambda)_*\mu_{\Lambda,\beta}$. A Borel probability measure μ_∞ on Ω_∞ is an *infinite-volume limit point* at coupling β if there are periodic lattices Λ_k with $\min_\lambda L_\lambda^{(k)} \to \infty$ such that $\int F\, d\tilde\mu_{\Lambda_k,\beta} \to \int F\, d\mu_\infty$ for every continuous $F : \Omega_\infty \to \mathbb{C}$. Closed paths, holonomies and rectangular loops (with arbitrary $R, T \ge 1$) in $\mathbb{Z}^d$ are defined as in Definition 2.3.

Corollary 7.2 (String tension)**.** *Let* $0 < \beta < \beta_0(d)$ *and let* z_0, M, r *be as in Theorem 6.2 with* $m(r) \ne 0$ *(e.g.* $G = SU(N)$ *and* $n(r) \ne 0$*).*

(i) For every periodic lattice with all $L_\lambda \ge 3$ *and every rectangular loop* C *with sides* R, T *in a plane* (μ,ν) *with* $L_\mu L_\nu \ge 2RT$*:* $\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \le d_r \exp(-\sigma(\beta)RT + \kappa(\beta)|C|)$*.*

(ii) For every infinite-volume limit point μ_∞ *and every rectangular loop* C *in* $\mathbb{Z}^d$ *with sides* $R, T \ge 1$*:* $\big|\int \chi_r(U_C)\, d\mu_\infty\big| \le d_r \exp(-\sigma(\beta)RT + \kappa(\beta)|C|)$*.*

(iii) Consequently, with $C_{R,T}$ *any rectangular loop with sides* R, T*,*

$$\liminf_{\min(R,T)\to\infty}\Big(-\frac{1}{RT}\log\sup_\Lambda\big|\langle \chi_r(U_{C_{R,T}})\rangle_{\Lambda,\beta}\big|\Big) \;\ge\; \sigma(\beta),$$
$$\liminf_{\min(R,T)\to\infty}\Big(-\frac{1}{RT}\log\Big|\int \chi_r(U_{C_{R,T}})\, d\mu_\infty\Big|\Big) \;\ge\; \sigma(\beta),$$

where the supremum is over all periodic lattices with all $L_\lambda \ge 3$*,* $L_\mu > R$*,* $L_\nu > T$ *and* $L_\mu L_\nu \ge 2RT$ *(with* $-\log 0 := +\infty$*). In words: the string tension in every representation of non-zero charge, defined by any of these limits, is at least* $\sigma(\beta)$*, uniformly in the volume.*

Proof. (i) $L_\mu L_\nu \ge 2RT$ gives $A = RT$ in Corollary 6.3. (ii) Let Λ_k be as in Definition 7.1. For k large, $L_\mu^{(k)} > R$, $L_\nu^{(k)} > T$, $L_\mu^{(k)}L_\nu^{(k)} \ge 2RT$ and all $L_\lambda^{(k)} \ge 3$; then the image of C in Λ_k is a rectangular loop $\bar C$ with the same sides, and $\chi_r((\iota_{\Lambda_k}U)_C) = \chi_r(U_{\bar C})$, so $\int \chi_r(U_C)\, d\tilde\mu_{\Lambda_k,\beta} = \langle \chi_r(U_{\bar C})\rangle_{\Lambda_k,\beta}$, which is bounded as in (i). Since $\tilde U \mapsto \chi_r(\tilde U_C)$ is continuous on Ω_∞, the bound passes to the limit. (iii) By (i) and (ii), both quantities in brackets are at least $\sigma(\beta) - (2\kappa(\beta)(R+T) + \log d_r)/(RT)$, which tends to $\sigma(\beta)$ as $\min(R,T) \to \infty$. □

Remark 7.3 (On infinite-volume limits). Infinite-volume limit points exist: for G a compact Lie group, Ω_∞ is compact and metrisable, so every sequence of probability measures on it has a weakly convergent subsequence (Prokhorov's theorem; e.g. [Bil99, Section 5]). They are invariant under translations of $\mathbb{Z}^d$, since each $\tilde{\mu}_{\Lambda,\beta}$ is ([F, Theorem 'Basic properties' (`thm:basic`)](iv)). Corollary 7.2 holds for all of them and does not use uniqueness. Uniqueness of the limit point (milestone T1.1 of the research program; cf. [F, Remark 'On the constants and on what this proves'], item (iii), the remark following `thm:strong`) is proved in Part II [II] for $\beta < \beta_*(d) = \operatorname{artanh} \frac{1}{6(d-1)}$, a range containing $(0, \beta_0(d))$.

8 Representations without central charge; two dimensions

Lemma 8.1 (Single-link vanishing). *Let C be a closed path, $r \neq 0$ irreducible and $X \subseteq \mathcal{P}$.*

(a) If a link ℓ occurs exactly once in C and $\ell \notin E(X)$, then $\Phi_{r,C}(X) = 0$.

(b) If a link $\ell \notin E_C$ lies in exactly one plaquette of X, then $\Phi_{r,C}(X) = 0$.

Proof. (a) After a cyclic shift of C (which does not change $\chi_r(U_C)$), $U_C = U_\ell^\epsilon B$ with $\epsilon = \pm 1$ and B independent of U_ℓ, and no factor $f_\beta(U_p)$, $p \in X$, depends on U_ℓ. By Fubini it suffices that $\int \chi_r(U^\epsilon B)\, dU = \operatorname{Tr}\big(\int D^r(U^\epsilon)\, dU\, D^r(B)\big) = 0$, which holds because $\int D^r_{ab}(U)\, dU = 0$ for $r \neq 0$ ([F, Cited Theorem `thm:pw`](ii) with the trivial representation) and dU is inversion invariant. (b) Let p be the plaquette of X containing ℓ. Then $U_p = A\, U_\ell^{\pm 1} B$ with A, B independent of U_ℓ, and no other factor of the integrand depends on U_ℓ; integrate U_ℓ first and use Lemma 2.2(c) (this is [F, Lemma 'Closed surfaces' (`lem:closed`)]). □

Proposition 8.2 (Perimeter bound for every non-trivial representation). *Let G, ρ be as in [F, Convention 'Group'], $0 < \beta < \beta_0(d)$, C a rectangular loop and $r \neq 0$ irreducible (no assumption on the centre). Then*

$$\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \leq \frac{d_r}{1 - x(\beta)} \exp\Big(-\Big[\frac{\sigma(\beta)}{4} - \kappa_1(\beta)\Big]|C|\Big).$$

For $d \geq 3$ the rate is positive for all $\beta < \beta_0(d)$: $\frac{\sigma}{4} - \kappa_1 > \frac{1}{4}\log 22 - \frac{3}{5} > 0.17$ if $d \geq 4$, and $\frac{\sigma}{4} - \kappa_1 > \frac{1}{4}\log 14 - 4(\log\frac{13}{12} + \frac{1}{13}) > 0.03$ if $d = 3$. For $d = 2$ and all $\beta < \beta_0(2)$,

$$\big|\langle \chi_r(U_C)\rangle_{\Lambda,\beta}\big| \leq \frac{d_r}{x(\beta)(1 - x(\beta))} \exp\Big(-\Big[\frac{\sigma(\beta)}{2} - \kappa_1(\beta)\Big]|C|\Big),$$

with $\frac{\sigma}{2} - \kappa_1 > \frac{1}{2}\log 6 - 2\log\frac{5}{4} - \frac{2}{5} > 0.049$. Hence in every dimension $d \geq 2$ and for all $\beta < \beta_0(d)$, rectangular Wilson loops in non-trivial representations decay at least exponentially in their length.

Proof. By Lemma 8.1(a), $\Phi_{r,C}(X) \neq 0$ forces $E_C \subseteq E(X)$, hence $|X| \geq |E(X)|/4 \geq |C|/4$. Apply Proposition 6.1 with $n_0 = \lceil |C|/4 \rceil$ and $t = 1/K$ as in the proof of Theorem 6.2, and use $x^{\lceil |C|/4\rceil} \leq e^{-\sigma|C|/4}$. By Lemma 2.8, $\sigma > \log(K+2)$ and $\theta < \frac{1}{K+1}$, so $\kappa_1 < 2(d-1)\big[\log(1 + \frac{1}{K}) + \frac{1}{K+1}\big]$. For $d \geq 4$ this gives $\frac{\sigma}{4} > \frac{1}{4}\log(8d-10) \geq \frac{1}{4}\log 22 > 0.77$ and, as in Lemma 2.8, $\kappa_1 < \frac{d-1}{2d-3} \leq \frac{3}{5}$. For $d = 3$ ($K = 12$): $\frac{1}{4}\log 14 > 0.6597$ and $4(\log\frac{13}{12} + \frac{1}{13}) < 0.6279$. For $d = 2$ ($K = 4$), Lemma 8.3 shows that Proposition 6.1 applies with $n_0 = A$, and $A \geq R + T - 1 = \frac{|C|}{2} - 1$ because $RT - (R + T - 1) = (R-1)(T-1) \geq 0$ and $L_0 L_1 - RT \geq (R+1)(T+1) - RT = R + T + 1$; with $t = 1/K$ as in the proof of Theorem 6.2 and $x < 1$ this gives $|\langle \chi_r(U_C)\rangle| \leq \frac{d_r}{1-x} x^A e^{\kappa_1|C|} \leq \frac{d_r}{1-x} x^{|C|/2-1} e^{\kappa_1|C|}$, which is the displayed bound; finally $\frac{1}{2}\log 6 > 0.8958$ and $2(\log\frac{5}{4} + \frac{1}{5}) < 0.8463$. (The rates in these bounds are the infima over $0 < \beta < \beta_0(d)$, approached as $\beta \uparrow \beta_0(d)$.) □

Lemma 8.3 (Contributing loop polymers in two dimensions)**.** *Let $d = 2$, let C be a rectangular loop, $r \neq 0$ irreducible, and $X \subseteq \mathcal{P}$ with $\Phi_{r,C}(X) \neq 0$. Then $X \in \{D,\ \mathcal{P}\setminus D,\ \mathcal{P}\}$; in particular $|X| \geq A$.*

Proof. For $d = 2$ every link lies in exactly two plaquettes (Lemma 2.2(d)). By Lemma 8.1, every link of E_C lies in at least one plaquette of X, and every link of $E(X) \setminus E_C$ lies in two plaquettes of X. Consider $\mathbf{x} := \sum_{p\in X} p \in \mathsf{C}_2(\Lambda;\mathbb{Z}_2)$. Since $[p : \ell] \equiv 1 \pmod 2$ for $\ell \in E(p)$, the coefficient of ℓ in $\partial\mathbf{x}$ is the number of plaquettes of X containing ℓ, modulo 2; so $\operatorname{supp}\partial\mathbf{x} \subseteq E_C$. By Lemma 3.2(i), $\partial\mathbf{x}$ is a 1-cycle. A 1-chain $c = \sum_{\ell\in E_C} a_\ell \ell$ with $\partial c = 0 \pmod 2$ has $a_\ell = a_{\ell'}$ for any two consecutive links of C (their common site lies on no other link of E_C), so $c \in \{0, [C]\}$. If $\partial\mathbf{x} = [C] = \partial D \pmod 2$, then $\mathbf{x} - D$ is a 2-cycle of the two-dimensional torus Λ, hence 0 or $\mathbf{1}$ (Lemma 3.3 with $\mathbb{A} = \mathbb{Z}_2$ and the directions $0, 1$ in place of $1, 2$), i.e. $X = D$ or $X = \mathcal{P} \setminus D$. If $\partial\mathbf{x} = 0$, then $\mathbf{x} \in \{0, \mathbf{1}\}$; and $X \neq \emptyset$ since $E_C \subseteq E(X)$. So $X = \mathcal{P}$. In all cases $|X| \geq \min(RT, L_0L_1 - RT) = A$. □

Proposition 8.4 (Two dimensions: every non-trivial representation)**.** *Let $d = 2$, G, ρ as in [F, Convention 'Group'], $0 < \beta < \beta_0(2)$, C a rectangular loop and $r \neq 0$ irreducible. Then $|\langle\chi_r(U_C)\rangle_{\Lambda,\beta}| \leq d_r \exp(-\sigma(\beta)A + \kappa(\beta)|C|)$. In particular this holds for the adjoint representation of $SU(N)$, and for every non-trivial representation of a group with trivial centre such as $SO(3)$.*

Proof. By Lemma 8.3, Proposition 6.1 applies with $n_0 = A$; conclude as in the proofs of Theorem 6.2 and Corollary 6.3. □

Remark 8.5 (Perimeter law for representations of N-ality zero; expected, not proved here). Let $G = SU(N)$, $d \geq 3$ and $r \neq 0$ with $n(r) = 0$ (e.g. the adjoint representation). Then the selection rule is void ($A_0(C) = 0$), and the only general constraint is Lemma 8.1, which leads to the perimeter bound of Proposition 8.2. The expected behaviour at small β is the *perimeter law*

$$c_1(\beta)\, e^{-c_2(\beta)|C|} \ \leq\ \langle\chi_r(U_C)\rangle_{\Lambda,\beta} \ \leq\ d_r\, e^{-c_3(\beta)|C|}, \qquad c_1, c_2, c_3 > 0,$$

uniformly in rectangular loops, so that the string tension of r vanishes ("screening of adjoint charges by gluons"). The reason is geometric. For $\lambda \notin \{\mu, \nu\}$, $L_\mu \geq R+3$ and $L_\nu \geq T+3$, let F be the union of the $2(R+T)+4$ unit cubes spanned by e_μ, e_ν, e_λ that lie in $\{x^0_\lambda \leq x_\lambda \leq x^0_\lambda + 1\}$ and project onto the unit squares outside the rectangle sharing at least a vertex with its boundary. Then F is a solid torus, its boundary ∂F is a closed surface of $4|C| + 16$ plaquettes, and each link of C lies in exactly two plaquettes of ∂F. (If $L_\mu = R + 2$ or $L_\nu = T + 2$, the ring closes up around the torus and F is a handlebody of genus two instead.) For $r = \mathrm{adj}$, assigning the fundamental representation to these plaquettes (oriented as the boundary of F), the representations meeting at every link contain the trivial representation in their tensor product ($\mathrm{adj} \otimes \rho \otimes \bar\rho$ on C, $\rho \otimes \bar\rho$ elsewhere), so terms of order $\beta^{4|C|+16}$ are not excluded by any selection rule, whereas a surface spanning C has at least A plaquettes, each carrying a factor $O(\beta)$. For other r of N-ality zero other representations must be placed on ∂F: for the $\mathbf{10}$ of $SU(3)$, $\mathbf{10} \otimes \mathbf{3} \otimes \bar{\mathbf{3}}$ contains no invariant, but $\mathbf{10} \otimes \mathbf{8} \otimes \mathbf{8}$ does, which gives terms of order $\beta^{2(4|C|+16)}$. Proving the lower bound requires showing that such terms are not cancelled, which the absolute-value bounds of this article cannot do; we leave it open. In $d = 2$ no such surfaces exist, and Proposition 8.4 shows that the area law holds for every $r \neq 0$: the distinction between zero and non-zero N-ality is a phenomenon of $d \geq 3$.

9 Discussion

Remark 9.1 (The $U(1)$ test). The proof of Theorem 6.2 uses only compactness, the bound $\|f_\beta\|_\infty \leq \epsilon(\beta)$ and a central element of finite order acting non-trivially in r; nothing non-

abelian enters. Accordingly Corollary 6.5 proves the same area law for $G = U(1)$, for *every* non-trivial representation (all of them have non-zero charge with respect to a suitable finite subgroup of the centre $U(1)$). On the other hand, for $G = U(1)$ in $d = 4$ the area law fails at large β, while in $d = 3$ it holds at all β [GM82] (as summarised in [Cha19b, Section 4] and [CNS25a, Section 1]). The four-dimensional statement is proved in [FS82] for the Villain action and asserted there to extend to Wilson's action (see also [Gut80]); the precise class of actions covered by these proofs is to be checked against the originals, which are marked "verify". Two conclusions follow.

(i) Strong-coupling confinement is *not* a non-abelian phenomenon: it is a property of the disordered (high-temperature) regime together with the centre symmetry, which forces every contributing surface to span the loop.

(ii) In the terminology of `docs/03-research-program.md` (§3.1, principle 2), the argument does not pass the $U(1)$ test: it applies verbatim to $U(1)_4$, where the large-β conclusion is false (for the Villain action, and by the extension asserted in [FS82] for Wilson's action). Hence no argument which uses only the ingredients listed above can give the area law at large β in four dimensions; a proof of confinement at all β for $SU(N)$ must use a genuinely non-abelian mechanism at an essential step. This is the same lesson as item (ii) of [F, Remark 'On the constants and on what this proves'] (the remark following `thm:strong`) for the Dobrushin proof of the mass gap.

Remark 9.2 (Comparison with Osterwalder–Seiler and Seiler). The area law at small β was derived at leading order of the strong-coupling expansion by Wilson [Wil74] and proved rigorously by Osterwalder and Seiler [OS78] (Theorem 5.1 there, according to the citation in the proof of [CNS25a, Theorem 1.2]), for planar loops and gauge groups with non-trivial centre such as $SU(N)$ and $U(N)$, with the extension to general loops given in Seiler's monograph [Sei82]; both statements are as summarised in [Cha19a, Section 3]. The work of Osterwalder and Seiler is based on the convergent strong-coupling (high-temperature) cluster expansion [SZZ23, Section 1]; [CNS25a, Remark 1.5] describes its regime as one in which even a single plaquette is rare. We have not been able to consult [OS78, Sei82] directly (they are marked "verify" in the bibliography), so the following comparison is with these descriptions.

(i) *Mechanism.* As far as can be judged from these descriptions, the mechanism is the same: centre symmetry forces the strong-coupling surfaces attached to the loop to span it. Here the topological step is isolated as an exact statement about $\mathbb{Z}_M$-chains (Proposition 3.7), proved with the projection chain map and the homology of the 2-torus; on a periodic lattice it produces the correct area $\min(RT, L_\mu L_\nu - RT)$, which includes the complementary surface wrapping the torus. The general form, Theorem 6.2, is the periodic-lattice analogue of the general-loop statement: the area is the minimal size of a $\mathbb{Z}_M$-valued 2-chain with boundary $-m(r)[C]$.

(ii) *Ratio of partition functions.* Instead of the full cluster expansion we use the inductive Lemma 5.4, whose hypothesis is the Kotecký–Preiss estimate of [F]; the article thus relies on no cited theorem beyond standard facts about compact groups.

(iii) *Constants and hypotheses.* All constants are explicit. In the normalisation of [F] the threshold $\beta_0(d)$ is independent of N; this agrees with the description of the Osterwalder–Seiler regime as $\tilde{\beta} \ll_d 1$ for the action written with the normalised trace in [CNS25a, Figure 1]. The hypothesis is a single central element of finite order acting non-trivially, which covers $SU(N)$, $U(N)$, $U(1)$ and finite groups with non-trivial centre uniformly.

(iv) *Beyond the strong-coupling regime of [OS78].* The area law has recently been extended, for large N, to $\beta \lesssim_d N$ for $U(N)$ [CNS25a] and to the 't Hooft regime $\beta \lesssim_d N^2$ for

$U(N)$ and $SU(N)$ (and $SO(2N)$) [CNS25b] (in the normalisation of [F]; these papers write the action with the unnormalised trace and with the 't Hooft scaling, respectively), using the master loop equation and, respectively, the Durhuus–Fröhlich criterion [DF80] verified by Bakry–Émery methods [SZZ23]. For fixed N and large β the problem is open (Remark 9.1).

Remark 9.3 (Constants, sharpness, improvements). (i) For the fundamental representation ρ of $SU(N)$ and $RT < L_\mu L_\nu - RT$, the leading term of Lemma 5.3 is $X = D$ (if $RT > L_\mu L_\nu - RT$ it is $X = \Sigma \setminus D$, and both contribute if the two areas are equal), and a standard computation with [F, Cited Theorem 'Haar measure, Peter–Weyl' (`thm:pw`)](iv) and the character expansion gives $\Phi_{\rho,C}(D) = d_\rho\, u_\rho(\beta)^{RT}$ with $u_\rho = c_\rho/(Nc_0)$, where $u_\rho(\beta) = \frac{\beta}{4} + O(\beta^3)$ for $N = 2$ and $u_\rho(\beta) = \frac{\beta}{2N^2} + O(\beta^2)$ for $N \geq 3$. The string tension suggested by this term is therefore $-\log u_\rho(\beta) = \log \frac{1}{\beta} + O(1)$; our lower bound $\sigma(\beta) = \log \frac{1}{\beta} - \log(2e^2K) + o(1)$ has the same leading behaviour, so $\sigma(\beta)/(-\log u_\rho(\beta)) \to 1$ as $\beta \downarrow 0$, but it is smaller by an additive constant, and it does not reflect the larger decay rates expected for representations r with $c_r(\beta) = O(\beta^k)$, $k \geq 2$. A matching upper bound on the string tension (a lower bound on the Wilson loop) is not proved here; a general lower bound of area type was given by Seiler using reflection positivity [Sei78] (as described in [Cha19a, Section 3]). (ii) The threshold $\beta_0(d)$ comes from the crude count of Lemma 2.2(f) in the Kotecký–Preiss condition. It can be improved by using that only polymers without free links contribute ([F, Lemma 'Closed surfaces' (`lem:closed`)]), and the area rate σ by counting only surfaces with boundary C (excess-area expansions), neither of which is pursued here. (iii) Corollary 6.6 shows the trade-off between the area rate and the perimeter coefficient controlled by the parameter t of Proposition 6.1.

Remark 9.4 (Status in the research program). Together with [F, Theorem 'Strong-coupling clustering and mass gap, uniform in the volume' (`thm:strong`)], the results of this article establish the area-law part of the strong-coupling endpoint of the program on the confinement side (lower bounds on Wilson loops, and the perimeter law for N-ality zero, are not proved here). The area law is listed under milestone T0.7 of `docs/03-research-program.md` ("Theorem modulo cited KP theorem"), although [F] does not prove it ([F, Remark `rem:further`]); it now has a self-contained proof, uniform in the volume and valid for every infinite-volume limit point of the periodic Wilson measures. The uniqueness of the infinite-volume limit (T1.1) is proved in Part II [II] for $\beta < \beta_*(d) := \operatorname{artanh} \frac{1}{6(d-1)}$, a range containing $(0, \beta_0(d))$; so for $\beta < \beta_0(d)$ Corollary 7.2(ii) concerns the unique infinite-volume Gibbs measure [II, Corollary 'Area law in infinite volume']. By Remark 9.1, the large-β problem is untouched by these methods.

Summary of results

Statement	Where	Status
$\mathbb{Z}_M$-area of a rectangle on a torus is $\min(RT, L_\mu L_\nu - RT)$	Prop. 3.7	Theorem
Centre selection rule; vanishing of small loop polymers	Lem. 4.3, Prop. 4.4	Theorem
Loops winding with non-zero charge vanish (all β)	Prop. 4.5	Theorem
Loop expansion; ratio bound; counting	Lem. 5.3, 5.4, 5.8	Theorem
Area law, general groups and closed paths	Thm. 6.2, Cor. 6.3	Theorem
Area law for $SU(N)$, $n(r) \neq 0$	Thm. 1.1 = Cor. 6.4	Theorem
Area law for $U(1)$ and $U(N)$	Cor. 6.5	Theorem
String tension $\geq \sigma(\beta)$, uniformly and in infinite volume	Cor. 7.2	Theorem
Perimeter decay of rectangular loops for all $r \neq 0$; area law for all $r \neq 0$ in $d = 2$	Prop. 8.2, 8.4	Theorem
Perimeter law for N-ality zero in $d \geq 3$	Rem. 8.5	open (expected)

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